

Cross-Distribution Identity-Document Fraud Detection: Diversity-Axis Ensembles of Double-Trace Consistency Detectors

The FREUID Challenge 2026: Technical Report (Team MobilityAI)

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Abstract

The FREUID Challenge 2026 requires a scalar fraud score in $[0, 1]$ for each identity-document image, ranked by the FREUID Score ($\text{AuDET} + \text{APCER}@1\%\text{BPCER}$; lower is better). The test data is split by design: the public split (about 5%) is born-digital, while the private split (about 95%) decides the ranking and mixes synthetic and physically captured images, including two document types absent from training. A single submission is scored on both splits, and the FREUID Score heavily penalizes a model that collapses on part of the data. We therefore submit one model built for the captured and unseen-type private split, not one tuned for the public leaderboard. The submitted model is a Double-Trace Consistency (DTC) capture ensemble: each detector adds a high-pass forensic branch to a pretrained RGB backbone and uses the trace as a global feature and as a document-type-agnostic patch-consistency signal. The main gain comes from ensembling decorrelated backbones across architecture and resolution: adding a RegNetY-160 to two ViT-B detectors lowers the held-out captured-proxy score from 0.384 to 0.227. A public-leaderboard-tuned ensemble reaches 0.00041 on the public split but is near chance on captured data, so we do not submit it. Training uses the competition data and public external datasets with self-supervised augmentation, never the test data; inference is a plain probability average in a pinned container that runs offline.

Competition: The FREUID Challenge 2026 (IJCAI-ECAI 2026), Kaggle `the-freuid-challenge-2026-ijcai-ecai` [11]

Kaggle team: MobilityAI

Code: <https://github.com/hoppery/freuid-challenge-2026>

1 Introduction

Identity-document fraud has several threat models: physical manipulation and re-capture, localized digital editing, and fully synthetic (GenAI) content. The FREUID Challenge [11] combines these into a single task. The model outputs $P(\text{fraudulent}) \in [0, 1]$ for each image, scored by the FREUID Score ($\text{AuDET} + \text{APCER}@1\%\text{BPCER}$; lower is better).

The challenge splits the test data in a deliberate way. The *public* split is about 5% of the test data. It is born-digital and covers the five training document types. The *private* split is about

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95% and decides the full ranking. It is drawn from a hybrid of synthetic and physically captured images, and it adds **two document types that are not in training**. A detector that only memorizes the born-digital appearance of the seen types generalizes poorly to the captured and unseen-type content. The organizers also caution against optimizing the public leaderboard [11].

A single submission is scored on both splits. We cannot produce different predictions for the two splits, and per-image routing is unsafe (Section 5.4). We therefore submit one model. Section 5.2 shows why we do not submit the public-optimized model: it drops to near chance on captured data. Because the private split carries captured and unseen-type content, and because the metric heavily penalizes a model that collapses on part of the data, we submit a model built for robustness (our **private-best** model). We select it on a captured proxy, never on private labels.

Contributions.

1. We show that a model optimized for the born-digital public leaderboard collapses on captured data (Section 5.2); a strong public score therefore need not carry over to the captured, unseen-type private split that decides the ranking.
2. We propose the Double-Trace Consistency detector (DTC), a detector whose patch-consistency signal does not depend on the document type (Section 2.2, Fig. 1).
3. We show a diversity-axis ensembling principle: combining decorrelated architectures and resolutions improves the submitted ensemble, with the architecture axis giving the larger gain (Sections 2.4, 5).

Related work. Identity-document forgery and presentation-attack detection is an active benchmark area. FantasyID [5] provides captured bona-fide cards with GenAI manipulations, close to this challenge’s threat model. For synthetic (GenAI) content, adapter-based detectors on frozen vision-language features generalize across generators [8]. Manipulation and re-capture leave low-level traces that semantic features miss. High-pass residual streams expose splicing and reprint artifacts. Central Difference Convolutions capture the gradient micro-texture used in face anti-spoofing [16]. We build on the residual-trace idea, but we summarize it as a *consistency* statistic. Self-blended images [12] synthesize manipulation traces without manipulation labels. We adopt this idea for our self-supervised consistency target and for the field-tamper augmentation. We use self-supervised ViT features (DINOv2 [9], ViT [3]) together with a convolutional family (RegNet [10]) through `timm` [14]. These families have different inductive biases, which we use for decorrelation. For robustness we use Fourier Domain Adaptation [15] to randomize low-frequency appearance. Our ensembling builds on deep ensembles [6]; we add evidence on which type of diversity helps here.

2 Method

2.1 Overview and problem setting

Let x be an image and $y \in \{0, 1\}$ its fraud label. Each detector produces a fraud logit $\ell(x)$ and a probability $\sigma(\ell(x)) \in [0, 1]$, where σ is the sigmoid; the submitted score is the mean of the detectors’ probabilities (Section 2.6). The FREUID training data is born-digital and covers the seen document types; we augment it with external captured data (Section 3). Evaluation uses a public split $\mathcal{D}_{\text{pub}}$ (born-digital) and a private split $\mathcal{D}_{\text{prv}}$ shifted in acquisition (captured vs. born-digital) and label space (two unseen document types). $\mathcal{D}_{\text{prv}}$ is not available, and it conflicts with $\mathcal{D}_{\text{pub}}$ (Section 5.2): a model tuned for one split is weak on the other. We therefore submit one model built for the private split (Section 2.5). Every detector uses the same feed-forward pipeline: decode $\rightarrow$ resize with ImageNet normalization $\rightarrow$ DTC detector $\rightarrow$ sigmoid.

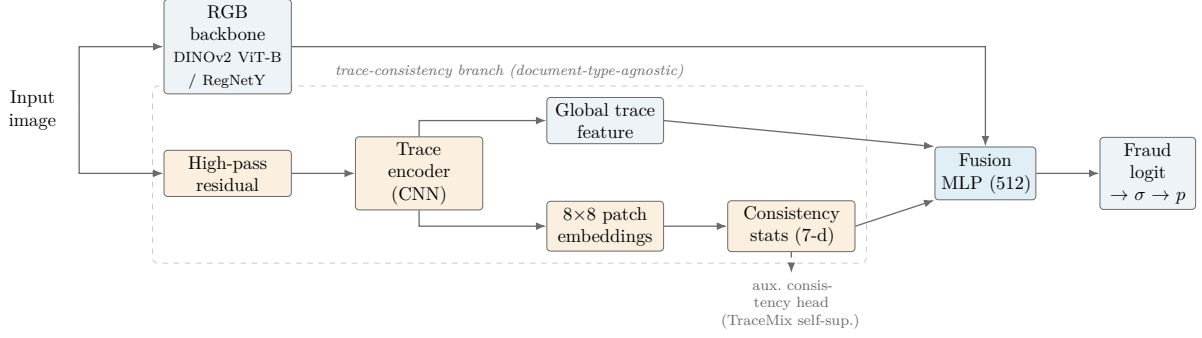


Figure 1: The Double-Trace Consistency detector (DTC). A pretrained RGB backbone is fused with a high-pass forensic branch. The patch-consistency statistics measure whether the acquisition trace is internally coherent. This signal does not depend on document-type appearance. Each deployed model is one such detector. The private-best model averages three of them (Section 2.4).

2.2 Double-Trace Consistency detector

Each detector (Fig. 1) combines an ImageNet/DINOv2-pretrained RGB backbone with a forensic branch. The forensic branch applies a Gaussian high-pass residual and a small strided CNN (**TraceEncoder**) that keeps the spatial layout. It produces (i) a global trace feature and (ii) an 8×8 grid of L2-normalized patch trace embeddings $\{z_i\}$. We take the off-diagonal pairwise cosine similarities of $\{z_i\}$ and compute a 7-dimensional **consistency statistic** $s(x)$ (mean, standard deviation, minimum, low percentiles, and per-patch max-similarity summaries). The name comes from the two uses of the high-pass trace: the global trace feature and the patch-level consistency signal $s(x)$. A design assumption is that a genuine capture, even a legitimately re-captured one, is uniformly traced, while a locally edited and reprinted forgery is not. Because $s(x)$ does not depend on the document type, an unseen-type genuine document should not be flagged merely for looking unfamiliar. The fraud logit $\ell(x)$ is an MLP over $[\text{RGB feature}; \text{global trace}; s(x)]$ (Linear $\rightarrow 512 \rightarrow \text{GELU} \rightarrow \text{Dropout}(0.1) \rightarrow \text{Linear} \rightarrow 1$). An auxiliary head on $s(x)$ is trained on self-supervised **TraceMix** coherent/incoherent labels, which are independent of the fraud label. The branch does not depend on the backbone, so any RGB family can host it. Training minimizes the binary cross-entropy on the fraud logit plus λ_{dte} times the same loss on the auxiliary consistency logit, with $\lambda_{\text{dte}} = 0.5$. We optimize with AdamW and a cosine schedule for three epochs in mixed precision; the ViT-B backbones are frozen except their last two blocks, and the RegNetY backbone is fully fine-tuned (Appendix A).

2.3 Capture robustness from data and augmentation

In our experiments, robustness to captured documents came from *data and augmentation* rather than from the architecture or loss changes we tried (Section 5.4). The private-best models train on FREUID plus diverse external captured corpora (Section 3) with heavy print-and-capture augmentation and FDA domain randomization [15]. With probability 0.4 we swap the low-frequency amplitude spectrum of an image with a donor of a *different* document type ($\beta \in [0.01, 0.08]$). This randomizes the type appearance and keeps the forensic phase.

2.4 Diversity-axis ensembling

A single capture detector is limited. The gain comes from averaging backbones that are *decorrelated* across two axes: input resolution and architecture family. The two ViT-B detectors at 896 and 1120 px are strongly correlated (Spearman $\rho = 0.79$ on the proxy). RegNetY-160 at 1120 px, a different architecture family, is less correlated with each ViT-B ($\rho = 0.69$ and 0.73) and gives the largest ensemble gain (Fig. 3).

2.5 The submitted model

We submit one model, the **private-best** ensemble: a probability average of three DTC detectors selected on the captured proxy, ViT-B/14 DINOv2 at 896 and 1120px and RegNetY-160 at 1120px. RegNetY-160 is the strongest single detector on the proxy (0.264) and the least correlated with the two ViT-B detectors, so it contributes the largest ensemble gain (Section 5.3). This is the model shipped in the reproducibility container.

As a reference only, we also build a **public-best** ensemble with a field-tamper augmentation. It reaches a low public score but collapses on captured data (Section 5.2), so we do not submit it.

2.6 Score calibration

Each detector outputs a fraud logit. A sigmoid maps it to a per-model probability. The submission score is the arithmetic mean of the ensemble probabilities. We apply no temperature or learned calibration, and no post-processing differs between train and test time. Unreadable images are assigned 0.5, so every input gives exactly one finite score in $[0, 1]$.

3 Data

We use the official FREUID `train/` images and `train.csv` metadata (Kaggle competition `the-freuid-challenge-2026-ijcai-ecai`). The private-best models also train on external data, combined with FREUID into a single manifest. Four corpora contribute *genuine* captured documents only: MIDV-500 [1], MIDV-2019 [2], MIDV-Holo [4], and BID [13] (Brazilian CNH/RG/CPF photographs). FantasyID [5] (Idiap), a dataset of fantasy ID cards, contributes both captured-genuine images and GenAI attack images (face-swap and text edits), so it also adds externally labeled fraud examples. All corpora are used under their public research licenses, and the pretrained weights under their upstream licenses.

Validation. The public leaderboard reflects the born-digital distribution, so it does not predict the captured private set (Section 5.2). For model selection we use a held-out FantasyID subset of 1,295 images (465 genuine captured cards and 830 GenAI-manipulated versions). It has no image overlap with the training data, so selection is not leaked. We use FantasyID because its captured genuine cards and non-training document styles match the two stated properties of the private test (Section 1). However, FantasyID is also a training source, so this is an *in-distribution* held-out estimate, not a cross-corpus test. It measures generalization to unseen FantasyID images and likely overestimates performance on a genuinely new corpus such as the private set. FantasyID is not an organizer-recommended proxy, and we do not claim it shares the private set’s distribution. We read its FREUID score as an indicative held-out number, not a lower bound. We also hold out a random 10% split of the training manifest for logging. The private set is not available at submission time, so proxy selection is the only option.

4 Inference

The container entrypoint indexes a flat directory of images. It runs the three checkpoints (build with `pretrained=False`, load embedded weights, forward pass, sigmoid) and writes the per-image probability average as `id,label1`. Test-time preprocessing is a plain resize to the native resolution with ImageNet normalization. The deployed models use no tiling, no multi-crop, and no test-time augmentation. The device is CUDA if available, else CPU. The per-model batch size follows the training config. Unreadable images are assigned 0.5. The exact command is in Section 6.

Table 1: Main results. FREUID Score $\downarrow$ (lower is better). We submit the private-best model; the public-best is a public-leaderboard reference we do not submit, because it collapses on captured data (Fig. 2). Captured-proxy figures are a small in-distribution held-out estimate.

Model	Public LB $\downarrow$	Captured proxy $\downarrow$
Private-best (submitted)	0.169	0.227
Public-best (reference, not submitted)	0.00041	≈ 0.99 (chance)

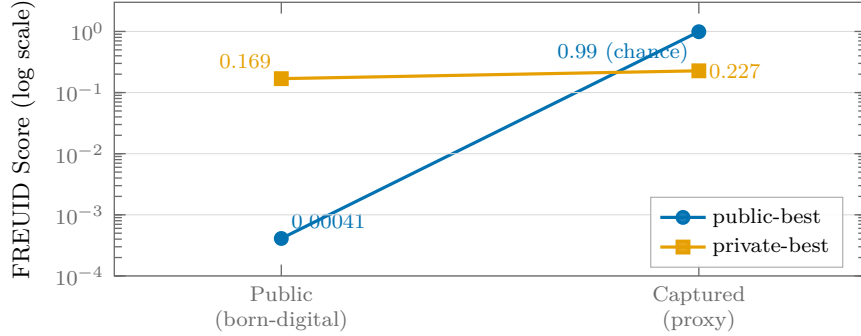


Figure 2: Each model fits only one distribution. A model is strong (low) on its own distribution and weak (high) on the other, so the two lines cross. Lower is better.

5 Results

5.1 Main results

Table 1 reports the submitted private-best model and the public-best reference on both splits. The private-best model reduces a two-ViT-B baseline from 0.384 to 0.227 on the captured proxy, and it stays well above chance on born-digital data. The public-best reference reaches a public FREUID Score of 0.00041 (with three models; 0.00039 for the full ensemble), but it collapses on captured data, so we do not submit it.

5.2 Public and captured performance

Fig. 2 plots each model on both distributions. The two lines cross. The public-best model is strong on born-digital data (0.00041) but at chance on captured data (about 0.99; a random classifier scores about 0.99 on APCER@1%BPCER). A likely reason is that the public-best model relies on born-digital forensic cues (fine editing traces, JPEG discontinuities) that print-and-capture re-acquisition overwrites, so they no longer separate genuine from fraud. The private-best model shows the opposite order. The gap between the two models is about 400 $\times$ on the public set, and the order fully reverses on the captured proxy. A single model therefore cannot serve both distributions well, and the public score does not predict captured performance. The private-best model stays well above chance on born-digital data (0.169), so it is our submitted model: it does not collapse on either split, and it is the stronger choice on the captured and unseen-type content that the private split contains and the public split does not.

5.3 Ensemble ladder (submitted model)

Fig. 3 shows the diversity-axis principle on the submitted model, measured on the captured proxy. A single ViT-B detector scores 0.427. Adding a second ViT-B at a different resolution (correlated) gives a modest gain to 0.384. Adding the decorrelated RegNetY-160 (a different architecture family) gives the large step to 0.227, the deployed three-model card.

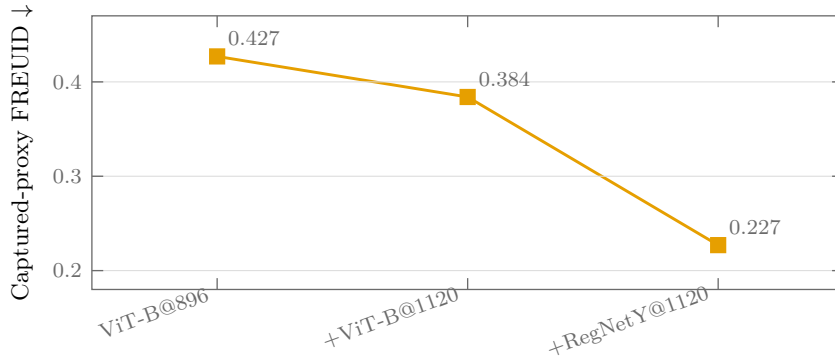


Figure 3: Ensemble ladder for the submitted three-model card (captured proxy, probability average). A second ViT-B at a new resolution gives a small gain; the decorrelated RegNetY-160 gives the large step to 0.227.

5.4 Rejected approaches

We also tested approaches that did not help, and we dropped them. Submitting the public-best ensemble on captured data was capture-blind (about chance). Adding MIDV-2020 to the capture mix gave no gain, because of a distribution mismatch. A learned domain router keyed on framing and context, and it misrouted cropped captured images. Warm-starting the capture model from the public-best checkpoint gave a worse peak than training from scratch. Architecture and loss levers on the capture axis (focal loss [7], CDC convolutions [16], MixStyle [17]) did worse than the data-diversity recipe.

6 Reproducibility

The reproducibility container ships the private-best model. The three checkpoints are fetched from a GitHub Release at build time (SHA-256 verified) and baked into the image, and the models build with `pretrained=False`. The container therefore runs with `-network none` (no runtime downloads) and writes only to `/submissions`.

- **Environment (pinned):** `torch==2.12.0` (CUDA 13), `timm==1.0.27`, `albumentations==2.0.8`, `opencv-python-headless==4.13.0.92`, `numpy==2.4.6`, `pandas==3.0.3`, `scipy==1.17.1`, `scikit-learn==1.9.0`, `pillow==12.2.0`, `pyyaml==6.0.3`, `pyarrow==24.0.0`; Python ≥ 3.11 .
- **Hardware / runtime:** training on $2 \times$ NVIDIA RTX PRO 6000 (Blackwell, 96 GB); inference uses a single GPU (about 1.6 h for the three models over the full test set on our hardware; CPU fallback works but is slow). The base image targets CUDA 13; on a host with an older driver, adjust the base tag and the `torch` index (weights are torch-version portable).
- **Randomness:** global seed 42. CUDA/cuDNN and AMP add minor non-determinism, within the proxy’s seed noise.
- **Repository:** <https://github.com/hopper/freuid-challenge-2026> (MIT license; the frozen commit SHA is in our reproducibility reply).

Build and run (organizer contract):

```
docker build -t freuid-repro:local .
docker run --gpus all --rm --network none \
  -v /path/to/flat/test/images:/data:ro \
  -v "$(pwd)/out":/submissions \
  freuid-repro:local # -> out/submission.csv (id,label)
```

7 Team and contributions

Both authors contributed to the method, the experiments, and the manuscript. Su-Yong An (corresponding author) led model development, training, and evaluation. Lae-Kyoung Lee contributed to the methodology, the analysis, and the manuscript.

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A Hyperparameters

Table 2: Per-model hyperparameters of the private-best ensemble. Shared: AdamW, weight decay 0.05, cosine LR schedule, 3 epochs, seed 42, AMP, DTC consistency weight $\lambda_{\text{dte}}=0.5$, `tracemix_p=0.5`, `fda_p=0.4`, heavy print-and-capture augmentation.

Hyperparameter	ViT-B @896	ViT-B @1120	RegNetY-160 @1120
Backbone	<code>vit_base_patch14_reg4_dinov2.lvd142m</code>		<code>regnety_160.tv2_in1k</code>
Image size	896	1120	1120
Batch size	24	12	8
Learning rate	1×10^{-4}	1×10^{-4}	5×10^{-5}
Weight decay	0.05	0.05	0.05
Epoch used	2	2	1
Backbone freezing	last 2 blocks + norm	last 2 blocks + norm	full fine-tune
Seed	42	42	42

B Additional ablations

See Fig. 3 (the diversity-axis contribution on the submitted model) and the rejected-approaches list in Section 5.4. Further ablations (capture-source diversity and resolution sweeps) are available in the repository notes.